Student Name: _____

Student Email: _____



Homework 1 Quiz (CS 7150)
Fall 2026

Instructions

- Write your name and your Northeastern email address on every page.
- You have 15 minutes. Write your answers in the spaces directly after each question.
- For worked questions, draw a single $\boxed{\text{BOX}}$ around your final answer, so that graders can tell the difference between your answer and your scratch work.
- Closed book: no notes, books, computers, calculators, or other resources, and no communication with anyone other than the proctor.
- You may not share or discuss these questions with any student who has not yet taken the quiz.

I attest that I have read and will follow the closed-book rules above.

Student signature: _____ Date: _____

1. McCulloch-Pitts neurons.

   Encode False as -1 and True as $+1$. A McCulloch-Pitts neuron computes $h = w_a a + w_b b + \beta$ and outputs $z = \text{sign}(h)$, for some choice of weights w_a, w_b and bias β.

   Fill in the last entry of the table below with the value of z (either $+1$ or -1) that makes it **impossible** for any single McCulloch-Pitts neuron to compute this function of a and b.

a	b	z
$+1$	$+1$	$+1$
$+1$	-1	-1
-1	$+1$	-1
-1	-1	$\boxed{}$

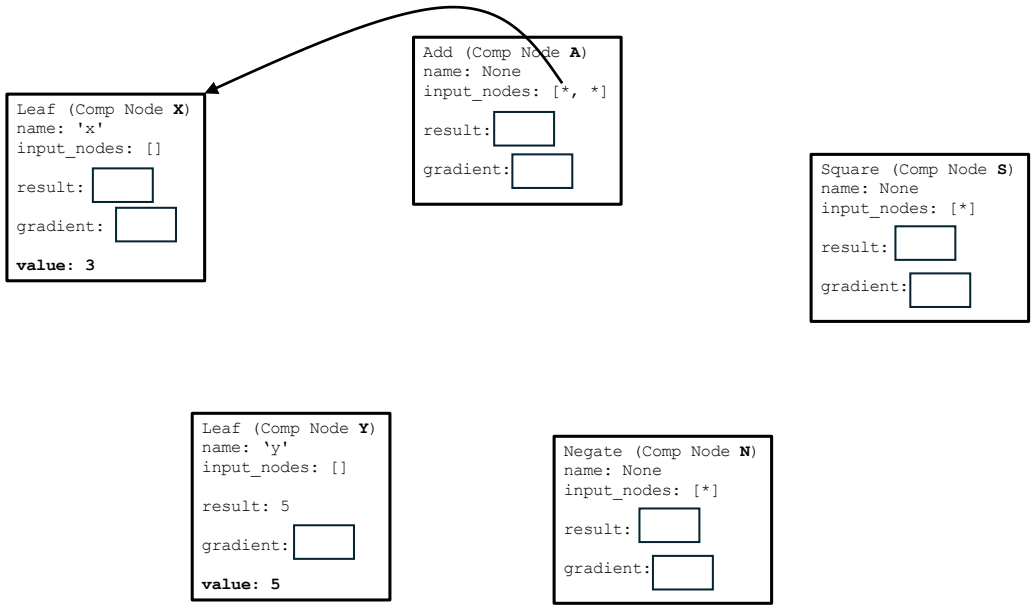
Solution: $z = +1$, giving XNOR (equivalently, "a equals b").

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2. Backpropagation.

The three-line program here builds a computation graph for $loss = (x + (-y))^2$ for specific values $x = 3$ and $y = 5$, and below is a partially-drawn diagram of the resulting graph, using the `ComputationNode` base class the homework. `Square(z)` is a `ComputationNode` that computes z^2. One arrow has already been drawn, indicating that node **A** refers to its first input node **X**; you should add the rest of the arrows and finish the diagram.

```
X = Leaf(3, name='x')
Y = Leaf(5, name='y')
loss = Square(X+(-Y))
```



(a) Draw the three remaining arrows needed to complete the graph, each pointing to a node referenced from another node's input list, in the same style as the arrow already drawn. Each arrow should start at a `*`.

Solution: N → Y, A → N, S → A.

(b) In the forward pass, which node's result is computed right after nodes X and Y have both their results computed? (Answer with one letter, A, N, or S) [ ]

Solution: N (it only needs Y, but must be computed before A, which needs both X and N).

(c) Backward pass: which node's gradient is computed right after S's? [ ]

Solution: A.

(d) The gradient computed at node X is $\frac{\partial \text{loss}}{\partial x}$. Write down the formula for the gradient computed at node N. [ ]

Solution: $\frac{\partial \text{loss}}{\partial(-y)}$, i.e., the gradient with respect to the value that node N computes, $-y$.

(e) Fill in all the values in the `result` boxes in the diagram above. One has already been done.

Solution: N.result $= -5$, A.result $= -2$, S.result $= 4$.

(f) Fill in all the numeric values in the `gradient` boxes in the diagram above, as computed by the `backward()` algorithm from the homework.

Solution: S.gradient $= 1$ (the loss's gradient with respect to itself); A.gradient $= 2A \cdot 1 = -4$; N.gradient $= 1 \cdot (-4) = -4$; X.gradient $= 1 \cdot (-4) = -4$; Y.gradient $= -1 \cdot (-4) = 4$.